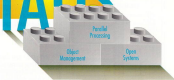


Steve Turner of TAO Systems explains the core concepts of this evolutionary environment to Parallelogram's Jerry Sanders.....



JS: You say that TAOS is designed as a high-performance, all-purpose operating system. Just how does it approach this goal?

ST: TAOS unifies the three most important developments in computing today - Parallel Processing, Object Management and Open Systems. Each is fulfilled in a novel, elegant manner, and combine in a natural evolution of computing development, as the figure below illustrates. The result is a big simplification of the system building process. Systems are flexibly assembled from interchangeable hardware and software components.



JS: How does TAOS manage to run the same binary code on any network of any type of 32 bit processor?

ST: The conceptual, and practical keystone is the "Virtual Processor" or VP. The VP is a generic 32 bit word CPU with 16 registers, R0 to R15. Software is built for the virtual processor, rather than a specific CPU.

The executable binary code for a virtual processor is called VPCode. At load time VPCode is translated, with negligible overhead, into optimised native code for the actual processor. The native code can then be cached. The end result is that VPCode performance is very close to native. VP Code instructions are normally dyadic (two operands). The first byte of each instruction defines the operation and the four addressing modes are constant, register, indexed indirect and double indirect).

JS: Have you had to invent a new instruction set for your virtual processor?

The VP Opcode Set

Specials:

ALL, BRK, ZAP, LITS, LITL, LITL, TAG, BRK, RM, RET, RZ, DL, FZD, DOP, IOP, FZL, LLD, DOL, FZXR, FZLL, FZOP, FZOD, CPS, CPH

Manipulates:

GOS, GOT, NEG, POP, PSR, TAO, SUB, SWP

Dyadics:

REM, ADD, AND, ASL, ASR, BOR, BIT, BSR, CMP, CPY, DFF, END, LEA, LSR, MELT, OR, SUB, SUB

ST: The VP Instruction set contains the standard operations required by a typical processor plus a small number of specific opcodes.

Since VP is a translated code, it includes pseudo-opcodes to perform some linker-like command functions necessary for translating the code efficiently into native machine code.

The linker type operations are TAG and LIT. The former sets a label at the current location, and the latter inserts a literal into the data output at that point. The ZAP operation is used to indicate to the Translator that the content of a register is no longer in use. This is then used in code optimisation. The monadic TAO operation transfers control to a specified kernel tool.

JS: Who would benefit from using TAOS as a development environment?

ST: Development for a "virtual machine" (an arbitrary network of virtual processors) is extremely productive. Even the embedded systems developer can benefit by using VPCode software for prototyping and maintenance. Software is developed in VP Assembly Code or using conventional languages which are compiled to VP Code. For the programmer the network of virtual processors can be completely transparent.

JS: In the embedded systems field, the virtual machine has to become real at some stage, when specific processors and topology are selected for the job. What happens then?

ST: At this point the programmer can choose to translate VPCode to native, and TAOS will then execute this directly.

JS: I can see that the Virtual Processor concept enables portability of binary software across arbitrary platforms. That's a very encouraging move in the direction of Open Systems.

ST: And there's more! The TAOS operating system itself uses the portability of VPCode to distribute processes dynamically across mixed networks, providing high-performance Parallel Processing. That's why we say that real Open Systems and Parallel Processing are inextricably linked.

A hands-on review of the TAOS system will appear in a forthcoming issue of Parallelogram

Two commercial parallel systems coming from IBM

RS/6000 workstation product is at the heart of IBM's client/server development program

New York, NY: IBM has announced a parallel processing computer server, the Power Visualisation System (PVS). Designed for visualising large supercomputer datasets, the system uses up to 32 first-generation Intel 860 processors in a non-uniform memory architecture.

At the front end is an RS/6000 Model 530 running AIX. The server can be shared by further networked RS/6000 workstations running Visualisation Data Explorer software which IBM has licensed from Silicon Graphics.

IBM did not use the latest 860hp processor in this system, according to a spokesman, because the company had not expected the processors to be available in time for the slated November 1991 ship date.

In Europe, IBM is involved in a major parallel processing collaboration, code-named Europa, with London-based Caplin Cybernetics, Shell Oil and the Edinburgh Parallel Computing Centre. The aim is to develop an even more powerful commercial massively parallel system.

This will leapfrog the 860hp to combine the 3rd generation 860 processor with the forthcoming Innos T9000 transputer, and will take full advantage of the unique features of both. The project has been funded from the United Kingdom Trade and Industry Department.

FAST FACTS No 134 IBM
SMP FACTS No 133 Europa

Silicon Graphics has introduced an ACE-compatible 3D graphics workstation, the Iris Indigo. "It's the hottest product we have ever announced and our competitors will be astounded by the number of

people buying it" said a spokesman.

Indigo uses the MIPS R3000A microprocessor, delivering 30 MIPS and 4.2 MFLOPS, and is priced from £6790 (diskless) to £8490 (£36Mb). It will run both Workstation and PC-style software: ports of WordPerfect, AutoCAD, Adobe Illustrator and Wings are under way.

Request Fast Fact No 100

ATG GigaLine of Toulouse, France has announced a 9 Gbyte optical WORM drive and is claiming the world record for 12 inch WORM drive capacity.

Request Fast Fact No 101

Storage Concepts of Irvine California, specialists in high performance and real time disk subsystems, has opened a UK office in High Wycombe, Bucks. Telephone 0494-642202.

Request Fast Fact No 102

Transtech Devices has pitched up its beach with Innos and has become a distributor of the latter's range of transputer modules (TRAMs).

Request Fast Fact No 103

Alsys Ltd has joined the Esprit SuperNode II project, and will build a super-fast Ada compiler for the next generation of transputer-based machines.

Request Fast Fact No 104

Parallel compilers specialists 3S Ltd is making some of its software project management expertise available to all with the launch of KeyTrack, a quality control and tracking system which runs on MS-DOS computers (no transputer required).

Request Fast Fact No 105

IN NEXT MONTH'S

ISSUE:

FDI Networking;
SCI Indigo Reviewed;
IBM Programming;
Transputers at the Post Office;
TA '91 Conference Report

TAOS

GENERAL PURPOSE PARALLEL OPERATING SYSTEM

AUTOMATIC PARALLEL PROCESSING

- Automatic network configuration
- Automatic process distribution
- Automatic load balancing
- Automatic optimised routing

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PORTABLE BINARY SOFTWARE

- No recompilation necessary for
- Changed network topology or size
- New processor types

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DEVELOPERS SYSTEM includes:

- Kernel, assembler, editor
- GUI, extensive libraries
- Hyperlink user interface builder

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Supports:

- Graphics Hardware by
- Innos, Paratech, Transtech, Quintek
- and B004/B008 Compatibles

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